PROJECT EVALUATION & ACADEMIC DOCUMENTATION GUIDE

CivicBrain

AI-Powered Geospatial Clustering & Municipal Grievance Redressal System

Domain: Full-Stack & Applied AI e-Governance	**Framework:** Digital India & MoHUA Smart Cities Mission
Evaluation: Lab Session & Capstone Portfolio	**Live URL:** https://civicbrain-nigam.vercel.app

CHAPTER 1: EXECUTIVE SUMMARY & PROBLEM STATEMENT

1.1 What is CivicBrain? (Conceptual Overview)

CivicBrain is an intelligent municipal e-Governance platform designed for city corporations and Nagar Nigams. It seamlessly connects citizens, field engineers, departmental supervisors, and municipal commissioners onto a single unified system. Whenever a citizen observes an urban defect (such as an open pothole, damaged streetlight, broken water pipeline, or waste overflow), they can file a report with photographic evidence and geographic location. The platform uses spatial algorithms to group duplicate reports within a 50-meter radius, routes the task to the appropriate department, enforces strict resolution countdown timers, and mandates before-and-after photographic verification before any complaint can be marked as resolved.

1.2 Real-World Problems Solved

- **Duplicate Report Floods:** In standard portals, dozens of citizens reporting the same physical hazard create multiple separate work orders. CivicBrain merges these into one master ticket while dynamically raising its urgency.
- **Unverified Task Closures:** Standard systems allow officers to close tickets without proof. CivicBrain enforces mandatory photographic evidence before closure.
- **Administrative Delays:** CivicBrain implements live countdown resolution timers that automatically escalate unresolved issues to senior supervisors.

CHAPTER 2: GLOBAL LITERATURE REVIEW & COMPARATIVE ANALYSIS

System Name	Region / Framework	Working Methodology	Key Limitation
NYC 311 / Boston 311	United States (Municipal)	Centralized citizen telephone and web ticketing.	No automated spatial deduplication; FIFO processing.
FixMyStreet	United Kingdom	Crowdsourced map reporting system.	Lacks photographic audit verification and dynamic penalty timers.
CPGRAMS & Swachhata	India (National MoHUA)	Departmental complaint routing portal.	Manual triaging overhead and no spatial proximity grouping.
DBSCAN Spatial Research	Academic Literature	Density-based spatial clustering algorithms.	High computation overhead; treats geographic space as flat 2D plane.

CHAPTER 3: THE 4 CRITICAL RESEARCH GAPS

Gap 1: Absence of Geospatial Deduplication

CRITICAL GAP

Issue: 30 citizens reporting the same broken streetlight creates 30 separate tickets, wasting municipal labor.

CivicBrain Solution: 50-meter spatial clustering algorithm that groups reports in the same category into one centralized Master Ticket.

Gap 2: Unverified Manual Closures

ACCOUNTABILITY

Issue: Workers close complaints from a dropdown without verifying that work was physically completed.

CivicBrain Solution: Dual-stage photographic audit requiring an After-Resolution photo before closure.

Gap 3: Static Priority Allocation

EFFICIENCY GAP

Issue: High-density community hazards receive the same urgency as isolated minor defects.

CivicBrain Solution: Dynamic priority escalation from Low to Medium, High, and Critical as more citizens report.

Gap 4: Network Fragility in Field

USABILITY GAP

Issue: Field workers operate in low-bandwidth zones where conventional web applications fail.

CivicBrain Solution: Resilient client-side caching maintaining operational continuity during network disruptions.

CHAPTER 4: MATHEMATICAL MODELING & ALGORITHMIC LOGIC

4.1 The Haversine Spatial Proximity Formulation

Because Earth is a curved sphere, standard flat-plane distance calculations produce significant metric distortion. CivicBrain applies the Spherical Haversine Equation:

Step 1: Difference in Latitude = Lat2 - Lat1 | Difference in Longitude = Lon2 - Lon1
Step 2: a = sin^2(Diff_Lat / 2) + cos(Lat1) * cos(Lat2) * sin^2(Diff_Lon / 2)
Step 3: Great-Circle Distance (d) = 2 * Earth_Radius * atan2(sqrt(a), sqrt(1 - a))
(Where Earth Radius R = 6,371 km, and d is calculated in physical kilometers)

Clustering Rules (Within 50 Meters):
- If calculated distance d is less than or equal to 0.05 km (50 meters) and category matches: Link to Master Ticket.
- Master Ticket affected citizen count increases by one (+1).
- If affected count reaches 5 or more: Priority automatically escalates to **CRITICAL**.

4.2 Dynamic Service Level Agreement (SLA) Decay Model

Remaining Time = Permitted Duration - (Current Time - Submission Time). When remaining time drops to zero, the ticket status shifts to Breached, and escalation notifications are triggered up the municipal hierarchy.

CHAPTER 5: FULL-STACK SYSTEM DESIGN & ARCHITECTURE

System Layer	Technologies Employed	Architectural Responsibility
Presentation Layer	React, TypeScript, Vite, Tailwind CSS	Role-based dashboard rendering, responsive mobile layouts, and national e-Governance UI styling.
Mapping Layer	Leaflet GIS, OpenStreetMap	Interactive GPS pin dropping, field officer route visualization, and commissioner density heatmaps.
Application Layer	Node.js, Express, Socket Events	Haversine spatial clustering execution, SLA state timers, and dual-factor email OTP dispatch.
Persistence Layer	MongoDB & Resilient Client Storage	Dual-tier data storage ensuring offline capability and zero data loss.
Hosting & Delivery	Vercel Production Edge CDN	Global high-availability distribution with Single Page Application routing rules.

CHAPTER 6: STAKEHOLDER PORTALS & USER JOURNEYS

1. Citizen Portal

CITIZEN

- **Pincode Auto-Location:** 6-digit postal pincode automatically resolves State, District, and Nagar Nigam.
- **Dual-Factor Security:** Real 6-digit verification code delivered directly to Gmail inbox.
- **Live 5-Stage Tracking:** Submitted → Clustered → Assigned → In Progress → Resolved.

2. Field Officer Dashboard

OFFICER

- **Prioritized Queue:** Tasks sorted by affected citizen density and urgency.
- **GPS Navigation:** Shortest route visualization between repair locations.
- **Photographic Verification:** Mandatory upload of an After-Resolution photo.

3. Department Head Dashboard SUPERVISOR

- **Live Countdown Timers:** Color-coded visual indicators for pending and expiring SLAs.
- **Workload Balancing:** Reassigning delayed tasks among available field staff.

4. Commissioner Dashboard EXECUTIVE

- **City-Wide Density Heatmap:** Real-time visual representation across all city wards.
- **Municipal Health Index:** Overall metric based on on-time resolution rates.

CHAPTER 7: REQUIREMENTS & AGILE USER STORIES

7.1 Key Requirements Overview

Functional: Accepts geo-tagged complaints with photos; auto-resolves addresses from postal pincodes; verifies email via OTP; automatically clusters duplicate reports within 50m; requires resolution proof photos before closing tickets.
Non-Functional: Spatial clustering calculation under 50 milliseconds; 99.9% uptime with offline resilience; 256-bit encryption; WCAG 2.0 AA accessibility compliance.

7.2 Agile User Stories with Story Points (SP)

User Story 1: Citizen Spatial Grievance Filing (5 Story Points) HIGH PRIORITY

As a Citizen, **I want to** submit a complaint with photo and GPS coordinates, **so that** the municipal corporation can fix the issue quickly. (*Acceptance Criteria:* Captures image and GPS; links nearby duplicates within 50m; issues tracking ID).

User Story 2: Field Officer Photo Verification (8 SP) | **User Story 3: SLA Countdown (5 SP)** CORE WORKFLOW

Officer: Uploads completion proof before closing ticket. **Head:** Views color-coded live timers (Green for normal, Yellow for warning, Red for breached).

CHAPTER 8: JIRA PROJECT MANAGEMENT & AGILE SCHEDULE

Sprint Iteration	Duration	Core Milestone Deliverables	Status
Sprint 1	Weeks 1 to 2	Architecture design, Government UI theme, Pincode API, Email OTP flow.	100% COMPLETED
Sprint 2	Weeks 3 to 4	Haversine spatial clustering engine, Master ticket linking, Citizen live tracker.	100% COMPLETED
Sprint 3	Weeks 5 to 6	Field Officer portal, GPS routing map, Before/After image upload, Offline sync.	100% COMPLETED
Sprint 4	Weeks 7 to 8	Commissioner heatmaps, SLA countdown engine, AI Chatbot, Vercel deployment.	100% COMPLETED

Project Role	Assigned Work Areas	JIRA Tracking Modules
Full-Stack Lead	System architecture, Frontend frameworks, Express backend, Cloud deployment.	CIVIC-101, CIVIC-104, CIVIC-201
Algorithm Specialist	Haversine proximity logic, SLA countdown decay, Email OTP security.	CIVIC-108, CIVIC-205, CIVIC-302
Frontend & GIS Engineer	Interactive GIS maps, Nagar Nigam theme, Mobile responsiveness, Chatbot.	CIVIC-112, CIVIC-308, CIVIC-401
QA & Documentation Lead	Literature review, research gap analysis, user stories, evaluation documentation.	CIVIC-405, CIVIC-501

CHAPTER 9: LAB EVALUATION & VIVA VOCE Q&A GUIDE

Q1: Why Haversine over Euclidean distance?

Ans: Euclidean distance treats Earth as flat, distorting real-world metrics. Haversine calculates great-circle distance over Earth's spherical curvature (Radius = 6,371 km), providing exact physical meter precision.

Q2: How to eliminate fake ticket closures?

Ans: Mandatory dual-stage photographic verification. An officer cannot close a ticket without uploading a live photograph of the completed repair, permanently audited on the ticket.

Q3: How is offline reliability maintained?

Ans: Client-side state caching. During network interruptions, complaints and user states are preserved in local storage and synchronized once connectivity resumes.

Q4: How does dynamic priority escalation work?

Ans: Clustered complaints increment an affected citizen counter on the Master Ticket. When 5 or more citizens report the same issue, priority automatically elevates to Critical.

Evaluation Summary: CivicBrain provides a robust, production-ready framework that fulfills all requirements across both research and full-stack development tracks, delivering measurable improvements in municipal efficiency and accountability.